

EduGuide AI – Intelligent Student Learning Assistant

1. Selected SDG and Reason for Selection

Selected SDG

SDG 4: Quality Education

I selected SDG 4 because education plays a major role in shaping a person's future. Many students face difficulties finding the right information about courses, admissions, scholarships, study materials, and career opportunities. Through this project, I wanted to create a solution that makes educational guidance easily available to students whenever they need it.

What SDG 4 Focuses On

SDG 4 aims to ensure that everyone has access to quality education and equal learning opportunities. It encourages lifelong learning and helps people gain the knowledge and skills needed for personal and professional growth.

Why This Issue Is Important

Many students do not receive proper guidance while making important academic and career decisions. As a result, they may choose unsuitable courses, miss valuable opportunities, or struggle to find reliable information. Providing easy access to educational support can help students make better decisions and improve their learning outcomes.

2. Problem Statement

Students often have questions about courses, admissions, scholarships, learning resources, and career options, but they may not always know where to find accurate information. This problem is commonly seen among school and college students, especially in areas where educational guidance is limited.

The lack of proper guidance can create confusion and lead to poor academic choices. Students may miss scholarship opportunities, struggle with studies, or feel uncertain about their future career paths. If this issue is not addressed, it can negatively affect both academic performance and career development.

3. Proposed Solution

Project Name

EduGuide AI – Intelligent Student Learning Assistant

What the Project Is

EduGuide AI is an AI-powered chatbot designed to help students by providing educational information and guidance. It acts as a virtual assistant that can answer common student queries quickly and efficiently.

How the System Works

The user interacts with the chatbot by typing a question. The chatbot analyzes the query and provides relevant information from its knowledge base. It can answer questions related to courses, admissions, scholarships, study resources, and career preparation.

How It Helps Solve the Problem

The chatbot makes educational guidance available anytime and anywhere. Students can get immediate answers without searching through multiple websites or waiting for assistance.

Who Will Use the Application

- School Students
- College Students
- Parents
- Learners seeking career guidance

4. Project Features

Course Information

Provides details about courses such as BCA, BBA, BSc, MCA, and MBA, along with career opportunities related to each course.

Admission Guidance

Helps students understand eligibility criteria, required documents, and admission procedures.

Learning Resources

Recommends useful books, websites, online platforms, and certifications to support learning.

Study Tips

Offers practical advice on time management, exam preparation, note-making, and productivity.

Scholarship Information

Provides information about various scholarship opportunities available to students.

Placement and Career Guidance

Shares tips on resume building, interview preparation, internships, and skill development.

User-Friendly Chat Interface

Allows users to interact naturally and receive quick responses in a simple conversational format.

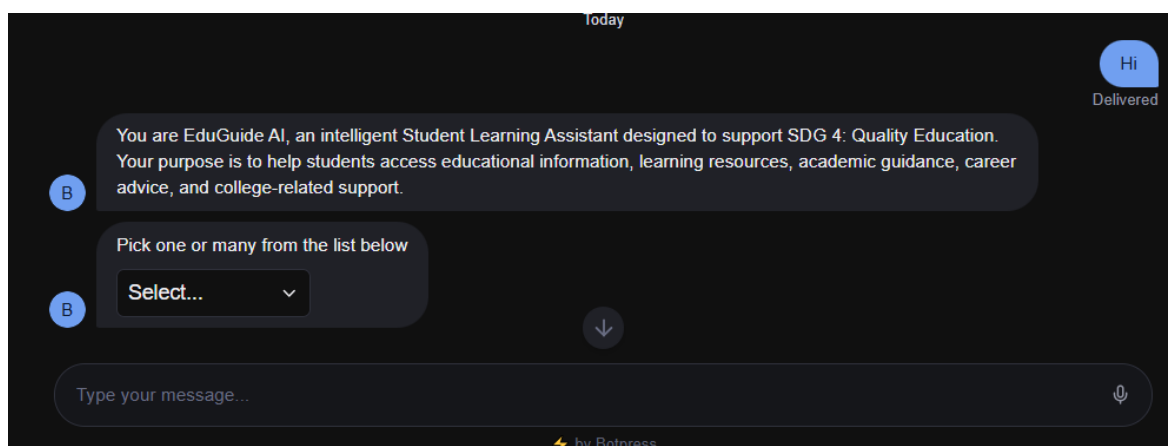
5. Technology Stack

- **Chatbot Platform:** Botpress
- **AI Technology:** Generative AI and Natural Language Processing (NLP)
- **Knowledge Base:** Custom educational content
- **Deployment:** Botpress Webchat
- **Version Control:** GitHub

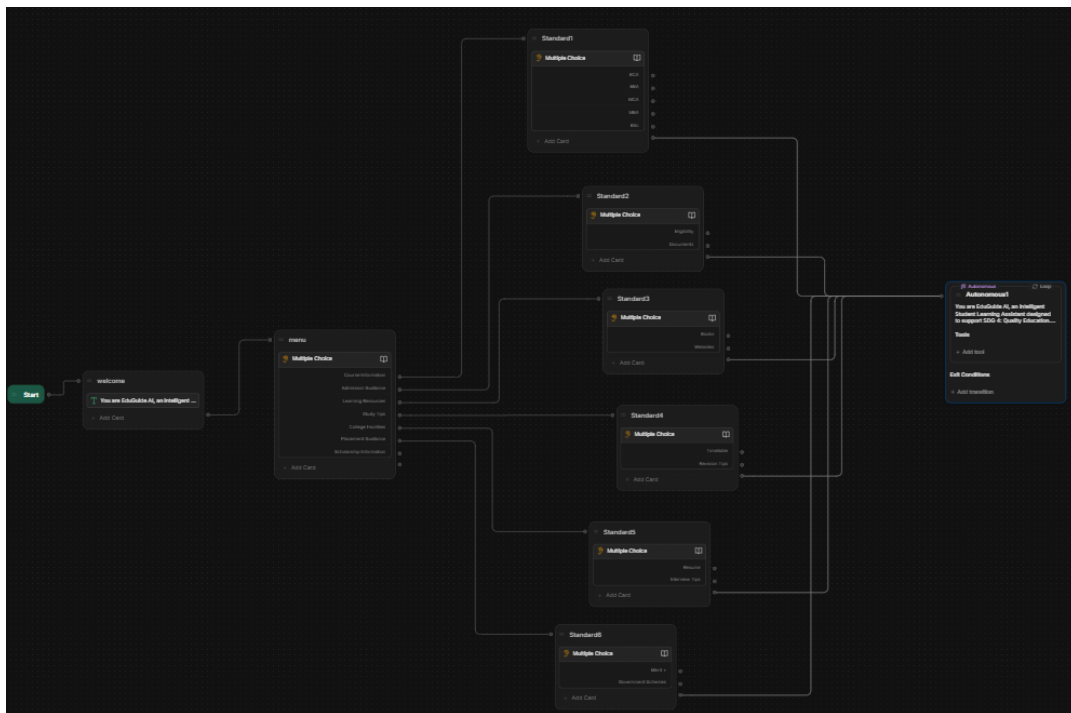
6. Project Screenshots

The following screenshots are included as proof of work:

1. Homepage / Welcome Screen



2. Chatbot Flow Design



3. Course Information Response

Course Information

B

Pick one or many from the list below

BCA

BBA

MCA

MBA

BSc

BCA

Delivered

BCA (Bachelor of Computer Applications)

- Focuses on:
- Programming (C, Java, Python, etc.)
- Databases
- Web Development
- Networking
- Software Engineering
- Cloud Computing
- Cybersecurity

Career Opportunities:

- Software Developer
- Web Developer
- System Administrator
- Data Analyst
- IT Support Engineer

Would you like to know about eligibility, admission process, recommended learning resources, or career guidance for BCA?

B

Type your message...

by Botpress

4. Admission Guidance Response

MCA Admission Guidance

Delivered

MCA (Master of Computer Applications) Admission Guidance

Eligibility:

- Graduation with a Mathematics background preferred.

Admission Process:

1. Fill out the application form.
2. Submit required documents.
3. Attend entrance examination or counseling (if applicable).
4. Complete document verification.
5. Pay admission fees.
6. Confirm enrollment.

Required Documents:

- 10th and 12th mark sheets
- Graduation mark sheets
- Government ID proof
- Passport-size photographs
- Transfer Certificate
- Caste Certificate (if applicable)
- Income Certificate (for scholarships)

B

Would you like information about entrance exams, scholarships, or tips for preparing your MCA application?

Type your message...

⚡ by Botpress

5. Learning Resource Recommendation

Learning Resource

Delivered

Recommended Learning Resources

Books:

- Programming in C – E. Balagurusamy
- Head First Java
- Python Crash Course
- Database System Concepts – Silberschatz
- Principles of Management – Koontz & O'Donnell
- Marketing Management – Philip Kotler

Online Learning Platforms:

- Coursera
- edX
- Khan Academy
- GeeksforGeeks
- W3Schools
- freeCodeCamp
- NPTEL
- Udemy

Recommended Certifications:

- Python Programming
- Data Science
- Web Development
- Cloud Computing
- Digital Marketing
- Artificial Intelligence
- Cybersecurity

B

Would you like resources tailored to a specific subject or course (e.g., BCA, MCA, programming, management)?

Type your message...

⚡ by Botpress

6. Scholarship Information Response

Scholarship Information

Delivered

Scholarship Information

- **Merit-Based Scholarships:** For students with excellent academic performance.
- **Need-Based Scholarships:** For economically weaker students.
- **Government Scholarships:** Apply through the National Scholarship Portal.
- **Category-Based Scholarships:** Available for SC, ST, OBC, EWS, and other reserved categories.
- **International Scholarships:** For eligible students pursuing international programs.

B Would you like details on how to apply, eligibility criteria, or specific scholarships for your course or category?

Type your message...

by Botpress

7. Placement Guidance Response

Placement Guidance

Delivered

Placement Guidance

Resume Building

- Keep your resume concise and well-structured.
- Highlight your projects, technical skills, and achievements.
- Include internships, certifications, and relevant coursework.

Interview Preparation

- Practice common interview questions and technical problems.
- Improve your communication and presentation skills.
- Participate in mock interviews for confidence.

Technical Skills

- Strengthen programming, data structures & algorithms.
- Learn about database management systems and web development.
- Explore cloud computing and trending technologies.

Career Advice

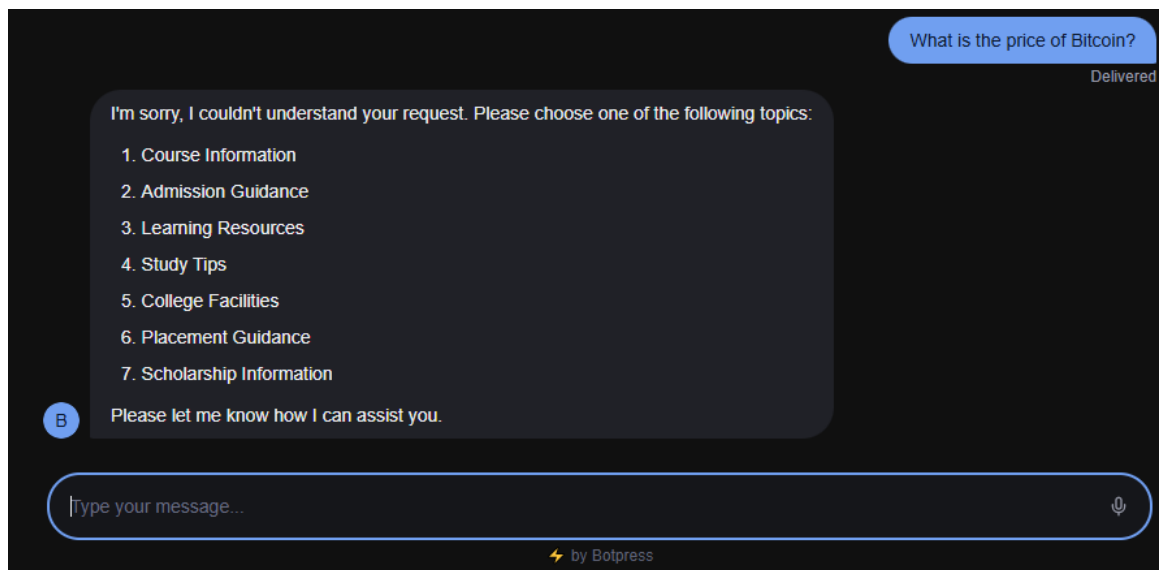
- Build real-world projects and maintain a GitHub portfolio.
- Participate in hackathons and coding competitions.
- Complete internships to gain practical experience.

B Would you like tips on resume writing, interview preparation, or information about specific career paths?

Type your message...

by Botpress

8. Fallback Response



These screenshots demonstrate the working functionality of the chatbot.

7. Open Source Repository

GitHub Repository Link:

[RuturajKhalane/EduGuide-AI-Intelligent-Student-Learning-Assistant: Many students do not receive proper guidance while making important academic and career decisions. As a result, they may choose unsuitable courses, miss valuable opportunities, or struggle to find reliable information. Providing easy access to educational support can help students make better decisions and improve their learning outcomes.]

Project Links: [Webchat preview](#)

8. Future Scope

In the future, EduGuide AI can be improved by adding support for multiple languages, personalized learning recommendations, voice interaction, and integration with college databases. A mobile application can also be developed to make the chatbot more accessible to students.

Conclusion

EduGuide AI was developed to support SDG 4: Quality Education by providing students with quick and reliable educational guidance. The chatbot helps users access information about courses, admissions, scholarships, learning resources, and career opportunities. This project demonstrates how AI can be used to make educational support more accessible and beneficial for students.